Fixed-Parikh Random Ordering and Quantitative LIFO Realization in $p{:}q$ Pushdown Normalization

Three Scales from Fixed Semantics to Physical LIFO Cost

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Abstract

Fix coprime positive integers p, q and let $m = p + q$. We study the centered real-time $p{:}q$ pushdown normalizer under a random-ordering experiment in which the semantic content is held fixed exactly: a word is chosen uniformly from the balanced fixed-Parikh fiber

$$\mathcal{F}_{gq,gp} = \{w \in \{0,1\}^{mg} : N_0(w) = gq,\ N_1(w) = gp\}.$$

Thus every sampled word has the same Parikh vector, the same zero endpoint residual, and the same shortest-completion semantics; only the order of the symbols varies. This isolates the physical cost created by LIFO realization.

We obtain three distinct operational scales. First, the words that incur no SEARCH or outer-WRITE obstruction admit an exact block factorization, giving exactly 2^g direct realizations and an explicit finite law for the first LIFO-obstruction time. In particular the first obstruction is $O_{\mathbb{P}}(1)$. Second, we introduce a recurrent block-boundary physical-lift chain and its excursion invariant measure ν. The semantic residual quotient assigns total invariant mass one to every residual shell, while physical chronology splits that mass among distinct stack/debt lifts. The total mass of empty-stack lifts,

$$\kappa_{p,q} = \sum_{d=-h}^{h} \nu(D_d, \varnothing), \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

is a $p{:}q$-specific physical rematerialization constant. If B_g denotes the number of bottom-WRITE events, then

$$\frac{B_g}{\sqrt{mg}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).$$

Third, away from a diffusive central residual band the normalizer has two finite-phase regenerative bulk regimes. For every bounded operational reward f, the total reward $C_g(f)$ differs in L^1 by only $O(\sqrt{n})$ from an affine function of the positive-residual occupation time. Since the occupation fraction of the limiting bridge is uniform, we obtain

$$\frac{C_g(f)}{n} \Longrightarrow v_-(f) + U\big(v_+(f) - v_-(f)\big), \qquad U \sim \mathrm{Unif}(0,1).$$

For the total LIFO-obstruction count $R_g = \#\mathrm{SEARCH} + \#\mathrm{OUTER\text{-}WRITE}$ this gives an explicit uniform limit law. Joint operational vectors have rank-one macroscopic covariance. We also prove a pathwise $p \leftrightarrow q$ physical-lift duality and, for the near-diagonal family $(q+1){:}q$, a universal Rayleigh-scale limit $\kappa_{q+1,q}/\sqrt{q(q+1)} \to 2$.

The probability tools used here—fixed-composition bridges, bridge occupation laws, recurrent-chain excursion measures, and Rayleigh local-time asymptotics—are classical. The contribution is their coupling to the shortest-demand quotient of the $p{:}q$ normalizer, which turns normalization chronology lost by the semantic quotient into an explicit, computable physical LIFO cost.

1 Introduction

For coprime positive integers p, q, write

$$m = p + q, \qquad \rho(w) = qN_1(w) - pN_0(w).$$

The weighted balance language is

$$L_{p,q} = \{w \in \{0,1\}^* : \rho(w) = 0\}.$$

A signed counter suffices to recognize this language, and deterministic one-counter computation is classical [1]. Our question is different. We work with a completion-aware pushdown normalizer in which the stack records a literal shortest completion demand, while bounded finite control stores the temporary debt created when LIFO order hides the next matching demand cell. Such a configuration is physically richer than its scalar residual: several state–stack configurations can project to the same shortest-demand semantics.

The operational starting point of this architecture is the completion-aware 2:1 deterministic pushdown prototype reproduced in Figure 1. In that machine, finite control and a binary work stack jointly encode the demand still required to restore weighted balance. Passing from that concrete 2:1 realization to the residual/shortest-demand viewpoint leads to the centered coprime $p{:}q$ normalizer used here. The figure is included to make this intellectual and operational provenance explicit; neither the 2:1 prototype nor the base $p{:}q$ architecture is a novelty claim of the present paper.

The present paper asks what quantitative information is lost by the semantic projection of that physical architecture. We hold the semantics fixed and randomize only chronology. Specifically, for an integer $g \geq 1$ we choose

$$W_g \sim \mathrm{Unif}(\mathcal{F}_{gq,gp}), \qquad \mathcal{F}_{gq,gp} = \{w : N_0(w) = gq,\ N_1(w) = gp\}.$$

Every word in this fiber has the same length $n = mg$, the same Parikh content, and the same endpoint residual zero. Thus any variation in SEARCH, WRITE, or debt is caused only by ordering.

This fixed-semantics experiment is deliberately complementary to Parikh-image questions, where symbol order is discarded by definition [2]. Quantitative pushdown models have also been studied through stack-cost objectives [5], but those objectives price stack contents along runs rather than compare LIFO chronology inside a fixed semantic fiber. Our aim is to isolate and quantify that chronology.

Base construction versus new results. The deterministic shortest-demand $p{:}q$ normalizer, its residual invariant, canonical-segment property, and exact WRITE normalization are the base automaton architecture for the present analysis; they are not novelty claims of this paper. We restate the needed definitions and short structural facts so that the probabilistic arguments are self-contained. The new results begin with the fixed-Parikh random-ordering experiment and quantify the physical chronology retained inside the fibers of the shortest-demand quotient.

Main result: three operational scales. Let

$$\begin{aligned} S_g &= \#\{\text{SEARCH transitions}\}, \\ O_g &= \#\{\text{outer-WRITE transitions}\}, \\ B_g &= \#\{\text{bottom-WRITE transitions}\}, \\ R_g &= S_g + O_g, \end{aligned}$$

and let τ_g be the first time at which SEARCH or outer-WRITE occurs. We prove the distributional trichotomy

$$\boxed{\tau_g = O_{\mathbb{P}}(1), \qquad B_g = \Theta_{\mathbb{P}}(\sqrt{n}), \qquad R_g = \Theta_{\mathbb{P}}(n).}$$

More precisely, τ_g has an exact finite survival law and a phase-geometric limit; $B_g/\sqrt{n}$ has a $p{:}q$-specific Rayleigh law; and R_g/n has an explicit uniform limit law when $p \neq q$.

The three scales arise from three distinct mechanisms. Direct realization fails after a bounded number of primitive blocks. Empty-stack rematerialization is a return/local-time phenomenon and therefore diffusive. Macroscopic SEARCH/outer-WRITE cost is controlled by the fraction of time the residual bridge spends in its positive and negative bulk regimes.

A physical-lift invariant. The central new quantity is

$$\kappa_{p,q} = \sum_{d=-h}^{h} \nu(D_d, \varnothing), \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor,$$

where ν is the recurrent excursion invariant measure of the block-boundary physical configuration chain. The semantic residual quotient has invariant mass one on each shell; $\kappa_{p,q}$ measures how much of the central rail is physically exposed with empty stack. It is the mean number of bottom-WRITE opportunities per primitive semantic zero-return excursion. The resulting limit law is

$$\frac{B_g}{\sqrt{n}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).$$

Thus a classical Rayleigh return mechanism acquires a nonclassical scale parameter determined by the physical lift structure of the pushdown normalizer.

Organization. Section 2 recalls the shortest-demand normalizer and fixes the operational observables. Section 3 identifies the fixed-Parikh model with a critical random-walk bridge. Section 4 gives the exact direct-realization and first-obstruction laws. Section 5 constructs the block-boundary physical-lift invariant measure and the constant $\kappa_{p,q}$. Section 6 proves the bottom-WRITE Rayleigh theorem. Section 7 derives the two exact bulk rates, and Section 8 proves the finite-to-macroscopic operational transfer theorem. Section 9 gives the uniform and rank-one macroscopic laws. Section 10 proves physical-lift duality, while Section 11 treats the near-diagonal family. Section 12 explains the finite matrix-kernel computation of $\kappa_{p,q}$. Computational checks and the novelty boundary appear in Sections 13 and 14.

2 The centered $p{:}q$ normalizer and operational observables

We recall only the portion of the centered normalizer needed in this paper. The presentation is self-contained.

2.1 Operational provenance: the completion-aware 2:1 prototype

The distinction between semantic demand and physical LIFO realization first appears concretely in the completion-aware 2:1 deterministic pushdown design shown in Figure 1. For

$$L_{2,1} = \{w : N_1(w) = 2N_0(w)\},$$

the center state C, side states L, R, and binary work stack jointly encode what must still be supplied to restore balance. A transition label $a, A/\gamma$ means that input a is consumed and stack top A is replaced by γ; the leftmost work symbol is the stack top and Z_0 is the bottom marker.

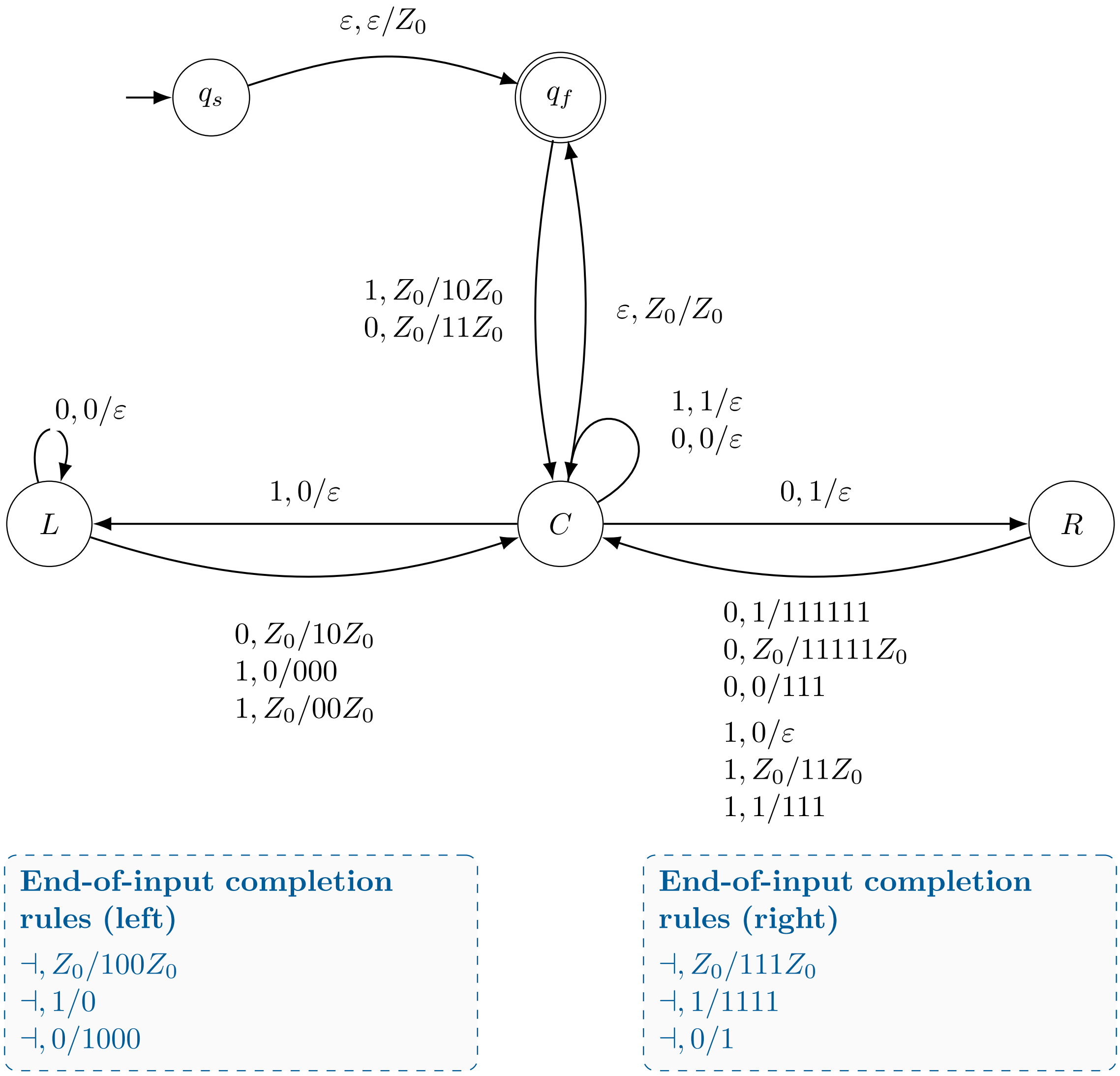

Figure 1: The completion-aware 2:1 DPDA that provides the operational provenance of the shortest-demand architecture, in the empty-word-included presentation. Solid arcs process binary input; dashed boxes list only right-endmarker completion operations. The figure records the concrete 2:1 starting point; the new results of the present paper begin only after passing to the general $p{:}q$ normalizer, fixing a semantic Parikh fiber, and randomizing symbol order within that fiber.

The dashed callout boxes are not ordinary ε-moves: they list completion operations enabled only when the right endmarker $\dashv$ is the next unread symbol.

Projecting away the initialization and acceptance wrapper leaves the semantic core $\{C, L, R\}$. If the physical stack is SZ_0, define

$$D(S) = 2N_0(S) - N_1(S), \qquad \Phi(C) = 0, \quad \Phi(L) = 3, \quad \Phi(R) = -3.$$

For every processed prefix the prototype satisfies

$$\rho_{2,1} = D(S) + \Phi(q_{\text{state}}), \qquad \rho_{2,1} = N_1 - 2N_0.$$

Thus part of a shortest completion demand may be carried by finite control even when the logical stack is empty. Forgetting that physical distribution leaves only the semantic residual and its canonical shortest completion. The centered $p{:}q$ construction below retains precisely this semantic/physical distinction, and the random-ordering experiment of this paper asks how much chronology remains visible in the physical lift after the semantic content has been fixed.

2.2 Shortest completion demand

For $r \in \mathbb{Z}$, let

$$\mu(r) = (x_r, y_r)$$

be the unique nonnegative solution of

$$px - qy = r$$

that minimizes $x + y$. All other nonnegative solutions are obtained by adding copies of the primitive zero-weight kernel

$$K = (q, p).$$

We use the canonical stack ordering

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases} \tag{1}$$

The leftmost symbol is the visible stack top.

Let

$$h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad D_{-h}, \ldots, D_h$$

be the centered debt states. For a logical stack word S define

$$W(S) = pN_0(S) - qN_1(S).$$

The state–stack residual invariant is

$$\rho = W(S) + dm \qquad \text{at configuration } (D_d, S). \tag{2}$$

The semantic projection reduces the physical stack plus virtual debt by complete kernel packets and returns exactly $\mu(\rho)$.

2.3 Transition classes

Suppose the current input is $a \in \{0, 1\}$. If $S \neq \varnothing$ and a equals the visible top, the machine performs a *MATCH*: it pops one cell and leaves d unchanged. If the visible top is 0 and $a = 1$, a mismatch increments debt while $d < h$; this is *SEARCH*. At $d = h$ the next such mismatch is an *outer WRITE*: the visible zero is removed, debt is reset to zero, and

$$U_+ = \eta(\mu((h+1)m))$$

is inserted before the surviving suffix. The negative rule is symmetric: input 0 over visible top 1 decrements debt while $d > -h$, and at $d = -h$ an outer WRITE inserts

$$U_- = \eta(\mu(-(h+1)m)).$$

If the logical stack is empty, the next symbol triggers a *bottom WRITE*. For every $|d| \leq h$,

$$(D_d, \varnothing) \xrightarrow{1} \big(D_0, \eta(\mu(dm + q))\big), \tag{3}$$

$$(D_d, \varnothing) \xrightarrow{0} \big(D_0, \eta(\mu(dm - p))\big). \tag{4}$$

Every WRITE therefore returns the physical stack to the exact canonical shortest-demand section for the new residual.

The construction has the following properties, all consequences of shortest-demand minimality and the centered debt bound.

Lemma 2.1 (WRITE frontier bound)**.** *Every canonical word created by a bottom or outer WRITE has shortest-demand vector* (x, y) *satisfying*

$$\min(x, y) \leq h.$$

For the outer-WRITE vectors $\mu(\pm(h+1)m)$*, the minority coordinate is at most* h *while the majority coordinate is at least* $h+1$*; in the positive case the minority one-count is* $< p$*, and in the negative case the minority zero-count is* $< q$*.*

Proof. For a bottom WRITE the target residual has the form

$$r = jm + q, \qquad -h-1 \leq j \leq h,$$

because $dm - p = (d-1)m + q$. Write $\mu(r) = (x, y)$ and $\ell = x + y$. Modulo m, the relation $px - qy = r \equiv q$ and $p \equiv -q \pmod{m}$ give

$$-q\ell \equiv q \pmod{m}.$$

Since $\gcd(q, m) = 1$, $\ell \equiv -1 \pmod{m}$, so $\ell = km - 1$. If both $x, y \geq h+1$, then $k \geq 2$. The minimum demand is kernel-free, hence either $x < q$ or $y < p$. In the first case $q \geq h+2$ and

$$r = m(x - qk) + q$$

forces $j \leq -q - 1 \leq -h - 3$, a contradiction. In the second case $p \geq h+2$ and

$$j = pk - y - 1 \geq p > h,$$

again a contradiction. Thus $\min(x, y) \leq h$.

For the positive outer WRITE, put $c = h+1$. The minimum solution is

$$\mu(cm) = (qk + c, pk - c), \qquad k = \left\lceil \frac{c}{p} \right\rceil.$$

Its second coordinate lies in $[0, h]$, is strictly less than p, and its first is at least $c = h+1$. The negative case is symmetric, with the zero-coordinate strictly less than q. □

Lemma 2.2 (Boundary guard)**.** *Suppose the current stack is a suffix of a canonical word whose minority count is at most* h*.*

(a) If the current state is D_h *and the stack begins with* 0*, then an outward* $1/0$ *mismatch can occur only when the surviving stack is unary zero.*

(b) If the current state is D_{-h} *and the stack begins with* 1*, then an outward* $0/1$ *mismatch can occur only when the surviving stack is unary one.*

Proof. Consider the positive case. Since the most recent WRITE, debt started at zero. Reaching D_h requires at least h previous positive mismatches, each of which popped a zero from the pre-existing canonical word. If the current visible zero were still in the mixed alternating part and a one survived later in the stack, then the h previously popped zeros would each have a distinct preceding one in the alternating part. Together with the current zero and a later surviving one, the original canonical word would contain at least $h+1$ zeros and at least $h+1$ ones, contradicting Lemma 2.1. Hence the surviving suffix is unary zero. The negative case is symmetric. □

Lemma 2.3 (Canonical-segment property)**.** *Immediately after every WRITE the machine is in state D_0 with stack $\eta(\mu(\rho))$, whose minority count is at most h. Until the next WRITE, every transition removes exactly one cell from that pre-existing canonical word. Hence every reachable logical stack is a suffix of the canonical word produced by the most recent WRITE.*

Proof. A bottom WRITE has the displayed canonical form by definition and satisfies Lemma 2.1. Between WRITEs, MATCH and SEARCH only remove the visible pre-existing cell. At an outer WRITE, Lemma 2.2 shows that the surviving suffix is unary. The newly inserted canonical word and that unary suffix concatenate to the canonical ordering of the new shortest demand; kernel-freeness follows from the minority bound in Lemma 2.1. Induction over WRITE episodes proves the claim. □

Lemma 2.4 (Zero-residual correctness)**.** *At the end of every processed binary prefix,*

$$\rho = 0 \quad \Longleftrightarrow \quad (D_d, S) = (D_0, \varnothing).$$

Proof. The reverse implication follows from (2). For the forward implication, consider the most recent WRITE. Immediately after it the stack was $\eta(\mu(r_0))$ for some residual r_0. Until the current time no further WRITE has occurred, so every intervening symbol has popped exactly one of those cells. If the current residual is zero, that intervening input is a completion of r_0. By minimality of $\mu(r_0)$, no completion can use fewer symbols than the length of the written stack. Since no more than that many cells can have been popped, equality is forced: the entire stack has been consumed. Equation (2) then gives $dm = 0$, hence $d = 0$. □

Operational observables. For a word w, let $S(w), O(w), B(w)$ be the numbers of SEARCH, outer-WRITE, and bottom-WRITE transitions. The primary LIFO-obstruction count is

$$R(w) = S(w) + O(w).$$

A bottom WRITE is not counted as an obstruction: it occurs when the physical stack is already empty and represents rematerialization rather than LIFO search.

3 The fixed-Parikh model as a critical bridge

For arbitrary $a, b \geq 0$ define the fixed-Parikh fiber

$$\mathcal{F}_{a,b} = \{w \in \{0,1\}^{a+b} : N_0(w) = a,\ N_1(w) = b\}.$$

Our balanced model is

$$W_g \sim \mathrm{Unif}(\mathcal{F}_{gq,gp}), \qquad n = mg.$$

Introduce i.i.d. symbols X_i under the critical law

$$\mathbb{P}(X_i = 1) = \frac{p}{m}, \qquad \mathbb{P}(X_i = 0) = \frac{q}{m},$$

and increments

$$\xi_i = \begin{cases} q, & X_i = 1, \\ -p, & X_i = 0. \end{cases}$$

Then

$$\mathbb{E}\xi_i = 0, \qquad \mathrm{Var}(\xi_i) = pq.$$

Write $\rho_t = \sum_{i \leq t} \xi_i$.

Lemma 3.1 (Canonical critical-bridge representation)**.** *For $n = mg$,*

$$\mathcal{L}(X_1, \ldots, X_n \mid \rho_n = 0) = \mathrm{Unif}(\mathcal{F}_{gq,gp}).$$

Proof. The condition $\rho_n = 0$ gives

$$mN_1(n) - pn = 0,$$

so $N_1(n) = pg$ and $N_0(n) = qg$. Every word with these counts has the same i.i.d. probability

$$\left(\frac{p}{m}\right)^{pg} \left(\frac{q}{m}\right)^{qg}.$$

Conditioning therefore makes the fiber uniform. □

At block boundaries set

$$Z_j = \frac{\rho_{mj}}{m}.$$

If K_j is the number of ones in the next m symbols, then under the critical i.i.d. law

$$Z_{j+1} - Z_j = K_j - p, \qquad K_j \sim \mathrm{Bin}\Big(m, \frac{p}{m}\Big). \tag{5}$$

Hence

$$\mathbb{E}(Z_{j+1} - Z_j) = 0, \qquad \mathrm{Var}(Z_{j+1} - Z_j) = \frac{pq}{m}. \tag{6}$$

The block skeleton is therefore a centered, irreducible, aperiodic, finite-range random walk on $\mathbb{Z}$.

4 Direct realization and the first LIFO obstruction

Call a word *direct* if no SEARCH or outer-WRITE transition occurs, equivalently if $R(w) = 0$.

4.1 Primitive direct blocks

The shortest demands after the first symbol are

$$\mu(-p) = (q-1, p), \qquad \mu(q) = (q, p-1).$$

Define

$$B_0 = 0\,\eta(q-1, p), \qquad B_1 = 1\,\eta(q, p-1). \tag{7}$$

Both words have length m and Parikh vector (q, p).

Theorem 4.1 (Direct-realization factorization)**.** *A balanced word $w \in \mathcal{F}_{gq,gp}$ satisfies $R(w) = 0$ if and only if*

$$w = B_{\varepsilon_1} B_{\varepsilon_2} \cdots B_{\varepsilon_g}$$

for a unique $(\varepsilon_1, \ldots, \varepsilon_g) \in \{0,1\}^g$. Consequently

$$\#\{w \in \mathcal{F}_{gq,gp} : R(w) = 0\} = 2^g \tag{8}$$

and

$$\mathbb{P}(R(W_g) = 0) = \frac{2^g}{\binom{mg}{qg}}. \tag{9}$$

Proof. Start from empty stack and zero debt. The first input symbol triggers a bottom WRITE. If it is 0, the newly written stack is $\eta(\mu(-p))$; if it is 1, the stack is $\eta(\mu(q))$. If no LIFO obstruction occurs thereafter, every subsequent input until the stack empties must match the visible top. Thus the entire first episode is forced to be B_0 or B_1. Each block has length m, consumes exactly (q,p) symbols, and returns the machine to $(D_0,\varnothing)$. Iteration gives the factorization. Conversely, each displayed block is realized by one bottom WRITE followed only by MATCH transitions, so any concatenation is direct. Uniqueness follows from the first symbol of each block. $\square$

By Stirling's formula,

$$\mathbb{P}(R(W_g)=0)\sim\sqrt{\frac{2\pi pqg}{m}}\left(\frac{2p^pq^q}{m^m}\right)^g. \tag{10}$$

The base

$$\lambda_{p,q}=\frac{2p^pq^q}{m^m}$$

is strictly less than one.

4.2 Exact first-obstruction law

Let

$$\tau_g=\min\{t\geq 1:\text{the transition at time } t \text{ is SEARCH or outer WRITE}\},$$

with $\tau_g=mg+1$ if no obstruction occurs. For $\varepsilon\in\{0,1\}$ and $0\leq s<m$, write

$$z_{\varepsilon,s}=N_0(\operatorname{pref}_s(B_\varepsilon)).$$

Theorem 4.2 (Exact first LIFO-obstruction survival law)**.** *For $0\leq k\leq g$,*

$$\mathbb{P}(\tau_g>km)=\frac{2^k\binom{m(g-k)}{q(g-k)}}{\binom{mg}{qg}}. \tag{11}$$

For $0\leq k<g$ and $1\leq s<m$,

$$\mathbb{P}(\tau_g>km+s)=\frac{2^k\sum_{\varepsilon=0}^{1}\binom{m(g-k)-s}{q(g-k)-z_{\varepsilon,s}}}{\binom{mg}{qg}}. \tag{12}$$

Proof. The event $\{\tau_g>km\}$ is exactly the event that the first k primitive blocks are direct. By Theorem 4.1, there are 2^k possible direct prefixes, each consuming (kq,kp) symbols. The remaining suffix is an arbitrary balanced fixed-Parikh word of size $g-k$, giving (11). For a partial block, after the first k complete direct blocks the next s symbols must equal $\operatorname{pref}_s(B_0)$ or $\operatorname{pref}_s(B_1)$. These two events are disjoint because B_0 and B_1 begin with different symbols. After fixing such a prefix, the remaining number of zeros is $q(g-k)-z_{\varepsilon,s}$, which gives (12). $\square$

Define

$$R_{p,q}(0)=1,$$

and, for $1\leq s<m$,

$$R_{p,q}(s)=\sum_{\varepsilon=0}^{1}\left(\frac{q}{m}\right)^{z_{\varepsilon,s}}\left(\frac{p}{m}\right)^{s-z_{\varepsilon,s}}.$$

Taking $g\to\infty$ with k,s fixed in Theorem 4.2 gives

$$\mathbb{P}(\tau_g>km+s)\longrightarrow\lambda_{p,q}^kR_{p,q}(s). \tag{13}$$

In particular,

$$\boxed{\tau_g=O_{\mathbb{P}}(1).} \tag{14}$$

4.3 Primitive balanced SEARCH distribution

At $g = 1$ no outer WRITE is required before return, and SEARCH occurs in mismatch pairs.

Proposition 4.3 (Primitive SEARCH law)**.** *On $\mathcal{F}_{q,p}$,*

$$S(W_1) \in \{0, 2, 4, \dots, 2\min(p, q)\},$$

and

$$\#\{w \in \mathcal{F}_{q,p} : S(w) = 2j\} = \binom{p}{j}\binom{q-1}{j} + \binom{q}{j}\binom{p-1}{j}. \tag{15}$$

Consequently

$$\mathbb{E}S(W_1) = \frac{2pq(m-2)}{m(m-1)}. \tag{16}$$

Proof. Condition on the first symbol. If it is 0, the bottom WRITE creates the deterministic target word $\eta(q-1, p)$, with $q-1$ zero-cells and p one-cells. The remaining input suffix has exactly the same Parikh vector. During this primitive episode no outer WRITE can occur: the debt at any prefix is the difference between the number of input ones and target ones seen so far, whose absolute value is at most $\min(p, q-1) \le h$. Thus every step pops one target cell. If exactly j target one-cells are hit by input zero, then conservation of the two symbol counts forces exactly j target zero-cells to be hit by input one, giving $2j$ SEARCH mismatches. The number of such suffixes is

$$\binom{p}{j}\binom{q-1}{j}.$$

If the first symbol is 1, the same argument with target $\eta(q, p-1)$ gives

$$\binom{q}{j}\binom{p-1}{j}.$$

Adding proves (15). For the mean, use

$$\sum_j j\binom{a}{j}\binom{b}{j} = \frac{ab}{a+b}\binom{a+b}{a}$$

in the two terms and divide by $\binom{m}{p}$. ☐

5 The block-boundary physical-lift chain

We now pass from exact finite obstruction counts to the recurrent physical structure that governs rematerialization.

Lemma 5.1 (Empty stack only at block boundaries)**.** *If the logical stack is empty after t processed symbols, then $m \mid t$.*

Proof. If $S_t = \varnothing$, the residual invariant (2) gives $\rho_t = dm$ for some integer d. On the other hand,

$$\rho_t = mN_1(t) - pt.$$

Hence $m \mid pt$. Since $\gcd(p, m) = \gcd(p, q) = 1$, we obtain $m \mid t$. ☐

Let X_j be the full physical configuration after jm symbols under the critical i.i.d. input law, and let

$$\pi(X_j) = Z_j = \rho_{jm}/m.$$

Let

$$o = (D_0, \varnothing).$$

By Lemma 2.4, $X_j = o$ exactly when $Z_j = 0$. Since the scalar block walk Z_j is recurrent, so is the reachable class of the physical block chain.

Let

$$T_o^+ = \min\{j \geq 1 : X_j = o\}.$$

Definition 5.2 (Physical-lift invariant measure)**.** For a reachable block-boundary physical configuration x, define

$$\nu(x) = \mathbb{E}_o \left[\sum_{j=0}^{T_o^+ - 1} \mathbf{1}_{\{X_j = x\}} \right]. \tag{17}$$

The standard excursion construction for an irreducible recurrent Markov chain makes ν an invariant measure normalized by $\nu(o) = 1$; see, for example, [11, Ch. 1].

Theorem 5.3 (Unit semantic-shell mass)**.** *For every $z \in \mathbb{Z}$,*

$$\boxed{\sum_{x:\pi(x)=z} \nu(x) = 1.} \tag{18}$$

Proof. Push ν forward under the factor map π. The result is an invariant measure for the irreducible recurrent random walk Z_j. Such an invariant measure is unique up to a multiplicative constant. Because the block walk is translation-invariant, counting measure on $\mathbb{Z}$ is invariant. At shell zero, Lemma 2.4 shows that the only physical state is o, so the pushed-forward mass equals $\nu(o) = 1$. The normalization therefore fixes the pushed-forward measure to be counting measure, proving (18). □

This is the quantitative form of the semantic/physical distinction: the residual quotient assigns one unit of invariant mass to each shell, while physical chronology determines how that unit is split among its lifts.

Definition 5.4 (Physical rematerialization constant)**.** For $-h \leq d \leq h$, let

$$E_d = (D_d, \varnothing), \qquad \theta_d^{p,q} = \nu(E_d),$$

and define

$$\boxed{\kappa_{p,q} = \sum_{d=-h}^{h} \theta_d^{p,q}.} \tag{19}$$

Because E_d belongs to semantic shell d, Theorem 5.3 gives

$$0 \leq \theta_d^{p,q} \leq 1, \qquad \theta_0^{p,q} = 1, \qquad 1 \leq \kappa_{p,q} \leq 2h + 1. \tag{20}$$

Let $\mathcal{E} = \{E_d : -h \leq d \leq h\}$ and consider the trace chain obtained by observing X_j only when it enters $\mathcal{E}$. Restriction of an invariant measure to a recurrent trace set is invariant for the trace chain. Therefore its stationary distribution is

$$\pi_d^{\mathcal{E}} = \frac{\theta_d^{p,q}}{\kappa_{p,q}},$$

and in particular

$$\boxed{\pi_0^{\mathcal{E}} = \frac{1}{\kappa_{p,q}}.} \tag{21}$$

6 Bottom-WRITE rematerialization has a Rayleigh law

The fixed-Parikh bridge can return to physical empty states only at block boundaries by Lemma 5.1. This allows an excursion decomposition at the block level.

A *primitive semantic excursion* is a block path beginning at o, avoiding o at positive intermediate block times, and ending at its first return to o. For such an excursion e, let

$$L(e) = \text{its length in blocks}$$

and let

$$R_B(e) = \#\{\text{empty-stack block states visited in } [0, L(e))\}.$$

Every such visit is followed, unless it is the terminal boundary of the full word, by exactly one bottom WRITE. Concatenating primitive semantic excursions therefore adds the rewards R_B exactly.

Lemma 6.1 (Excursion reward mean and exponential moments)**.** *Under the critical i.i.d. law,*

$$\mathbb{E}R_B(e) = \kappa_{p,q}. \tag{22}$$

Moreover there exists $s > 1$ such that

$$\mathbb{E}s^{R_B(e)} < \infty. \tag{23}$$

Proof. Equation (22) is the excursion formula (17) summed over the finite empty-state set $\mathcal{E}$. For (23), consider the finite trace chain on reachable empty states. From every such state there is a positive-probability finite input continuation that completes its residual and reaches E_0. Finiteness gives constants $L < \infty$ and $\varepsilon > 0$ such that from every trace state the probability of hitting E_0 within at most L trace transitions is at least ε. Hence the trace hitting time has a geometric tail, which implies an exponential moment for $R_B(e)$. $\square$

Let

$$b_g = \mathbb{P}(Z_g = 0) = \binom{mg}{pg}\left(\frac{p}{m}\right)^{pg}\left(\frac{q}{m}\right)^{qg}.$$

Define the bridge return generating function

$$G(z,1) = \sum_{g\geq 0} b_g z^g.$$

Let $F(z,u)$ be the probability generating function of primitive semantic excursions, with z marking block length and u marking R_B. Unique decomposition into primitive excursions gives

$$G(z,u) = \frac{1}{1 - F(z,u)}. \tag{24}$$

At $u = 1$, Stirling's formula yields

$$b_g \sim \sqrt{\frac{m}{2\pi pq}}\, g^{-1/2}. \tag{25}$$

A standard Abelian/Tauberian transfer therefore gives

$$G(z,1) \sim \sqrt{\frac{m}{2pq}}(1-z)^{-1/2} \qquad (z \uparrow 1), \tag{26}$$

and hence

$$1 - F(z,1) \sim \sqrt{\frac{2pq}{m}}\sqrt{1-z}. \tag{27}$$

The same square-root singularity is a standard source of Rayleigh local-time laws for zero-drift lattice-path bridges; compare [9]. We use only the elementary regular-variation consequences of this square-root behavior below. Here the reward is not scalar zero-local-time itself but the physical empty-lift reward carried by each primitive semantic excursion.

Theorem 6.2 (Bottom-WRITE Rayleigh law)**.** *Let $B_g = B(W_g)$. Then*

$$\boxed{\frac{B_g}{\sqrt{mg}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right).} \tag{28}$$

Equivalently, for every fixed integer $r \geq 1$,

$$\mathbb{E}\left[\left(\frac{B_g}{\sqrt{mg}}\right)^r\right] \longrightarrow \left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right)^r 2^{r/2}\Gamma\left(1+\frac{r}{2}\right). \tag{29}$$

Proof. Write

$$H_j(z) = \partial_u^j F(z,u)\big|_{u=1}.$$

By Lemma 6.1, $H_j(1) < \infty$ for every fixed j, and

$$H_1(1) = \kappa_{p,q}.$$

Differentiating (24) r times in u, the term with r factors of H_1 is

$$r!H_1(z)^r G(z,1)^{r+1}. \tag{30}$$

Every other Faà di Bruno term contains at most $r-1$ derivative factors of F and hence at most $G(z,1)^r$. More precisely, each such term is a product of some $H_{j_1}(z), \ldots, H_{j_k}(z)$ with $k \leq r-1$ and a factor $G(z,1)^{k+1}$. Since every H_j has summable nonnegative coefficients by Lemma 6.1, the convolution theorem for summable sequences against regularly varying coefficients shows that its coefficient is $O(g^{(k-1)/2}) = o(g^{(r-1)/2})$. Thus only (30) contributes at leading order.

Using (26), Karamata coefficient asymptotics, and the same standard convolution lemma (equivalently, the corresponding singularity-transfer calculation; cf. [12]),

$$[z^g]\partial_u^r G(z,1) \sim \frac{r!\kappa_{p,q}^r}{\Gamma((r+1)/2)}\left(\frac{m}{2pq}\right)^{(r+1)/2} g^{(r-1)/2}.$$

Dividing by b_g in (25) gives the conditional factorial moment

$$\mathbb{E}[(B_g)_r] \sim \kappa_{p,q}^r \left(\frac{m}{pq}\right)^{r/2} 2^{r/2}\Gamma\left(1+\frac{r}{2}\right) g^{r/2},$$

where the duplication formula for the Gamma function was used. Ordinary moments have the same leading term. These are exactly the moments of a Rayleigh variable with scale $\kappa_{p,q}\sqrt{m/(pq)}$ for $B_g/\sqrt{g}$, or equivalently scale $\kappa_{p,q}/\sqrt{pq}$ for $B_g/\sqrt{mg}$. The Rayleigh distribution is moment-determinate, so the method of moments proves (28). □

Corollary 6.3 (Mean and variance of rematerialization)**.** *With $n = mg$,*

$$\mathbb{E}B_g \sim \kappa_{p,q}\sqrt{\frac{\pi}{2pq}}\sqrt{n}, \tag{31}$$

$$\mathrm{Var}(B_g) \sim \kappa_{p,q}^2 \frac{4-\pi}{2pq}\, n. \tag{32}$$

7 Two regenerative bulk regimes

We now analyze the n-scale LIFO cost. Put

$$c = h + 1, \qquad k_+ = \left\lceil \frac{c}{p} \right\rceil, \qquad k_- = \left\lceil \frac{c}{q} \right\rceil.$$

The canonical outer-WRITE demands are

$$\mu(cm) = (qk_+ + c,\ pk_+ - c), \tag{33}$$
$$\mu(-cm) = (qk_- - c,\ pk_- + c). \tag{34}$$

To define the bulk constants, use the translation-invariant positive-tail extension obtained by quotienting complete unary packets (equivalently, adjoin an unbounded unary zero tail and retain the same finite phase dynamics). A finite deep-positive interval in the actual normalizer follows this extension until it hits the central boundary; only its first and last regeneration cycles can be truncated, and those endpoint pieces are handled in Section 8. In the homogeneous positive extension, consider the interval from one positive outer WRITE to the next. The mixed head of the newly written word is finite, while the surviving tail is unary zero. The next positive outer WRITE occurs after exactly pk_+ input ones have been read. Under the critical i.i.d. law the cycle length is therefore

$$T_+ \sim \mathrm{NB}\Big(pk_+, \frac{p}{m}\Big), \tag{35}$$

where T_+ counts trials through the final success. Similarly,

$$T_- \sim \mathrm{NB}\Big(qk_-, \frac{q}{m}\Big). \tag{36}$$

Thus

$$\mathbb{E}T_+ = mk_+, \qquad \mathrm{Var}(T_+) = \frac{k_+qm}{p}, \tag{37}$$
$$\mathbb{E}T_- = mk_-, \qquad \mathrm{Var}(T_-) = \frac{k_-pm}{q}. \tag{38}$$

In particular the outer-WRITE rates are

$$o_+ = \frac{1}{mk_+}, \qquad o_- = \frac{1}{mk_-}. \tag{39}$$

Theorem 7.1 (Exact bulk LIFO-obstruction rates)**.** *For $R = S + O$, the critical i.i.d. positive- and negative-bulk rates are*

$$\boxed{r_+ = \frac{2pq}{m^2} + \frac{(p-q)c}{m^2k_+},} \tag{40}$$
$$\boxed{r_- = \frac{2pq}{m^2} - \frac{(p-q)c}{m^2k_-}.} \tag{41}$$

Hence the SEARCH rates are

$$s_+ = r_+ - o_+, \qquad s_- = r_- - o_-. \tag{42}$$

Proof. We prove the positive case. Write

$$(x, y) = \mu(cm) = (qk_+ + c, pk_+ - c).$$

A positive cycle ends at the pk_+-th input one. All y one-cells in the canonical head are consumed before the outer boundary can be reached, by Lemma 2.2. Let M be the number of those one-cells that are consumed by input zero; each such event is a negative-direction SEARCH mismatch. Since these cells are reached before stopping and the input is i.i.d.,

$$\mathbb{E}M = y\frac{q}{m}.$$

Among the pk_+ input ones in the cycle, exactly $y - M$ match one-cells. Therefore the number of one-over-zero mismatches is

$$pk_+ - (y - M) = c + M.$$

The total number of LIFO obstructions in the cycle is consequently

$$c + 2M,$$

with mean

$$c + 2(pk_+ - c)\frac{q}{m}.$$

Dividing by the mean cycle length mk_+ gives (40). The negative calculation is obtained by interchanging zero and one, p and q, and reversing the debt sign; this gives (41). □

The difference

$$r_+ - r_- = \frac{(p-q)c}{m^2}\left(\frac{1}{k_+} + \frac{1}{k_-}\right) \tag{43}$$

is nonzero whenever $p \neq q$. Since coprime $p = q$ forces $p = q = 1$, the only symmetric ratio is 1:1.

8 Finite-to-macroscopic operational transfer

Let f_t be a bounded operational reward determined by the physical transition at time t; examples include indicators of SEARCH, outer WRITE, MATCH, or bounded functions of the debt state. Put

$$C_g(f) = \sum_{t=1}^{n} f_t, \qquad n = mg.$$

The positive-residual occupation count is

$$A_g = \#\{1 \le t \le n : \rho_t > 0\}.$$

Let $v_+(f)$ and $v_-(f)$ denote the corresponding critical i.i.d. positive- and negative-bulk reward rates. For R, these are r_+ and r_- from Theorem 7.1.

The proof uses a symbol-level finite-phase quotient in the two deep tails. We record the required periodicity explicitly because the rewards in this section are accumulated one input symbol at a time.

Lemma 8.1 (Symbol-level affine tail periodicity)**.** *For every $r \ge 0$,*

$$\mu(r + pm) = \mu(r) + (m, 0), \tag{44}$$

and for every $r \le 0$,

$$\mu(r - qm) = \mu(r) + (0, m). \tag{45}$$

Consequently, after quotienting complete unary m-packets, the reachable configurations in each sufficiently deep positive or negative residual tail have a finite symbol-level phase space.

Proof. For $r \geq 0$, the minimum solution $\mu(r) = (x, y)$ has $0 \leq y < p$: among the p residue classes for y, exactly one makes $r + qy$ divisible by p, and the resulting x is nonnegative. Hence $(x + m, y)$ is a nonnegative solution of weight $r + pm$ with the same $y < p$, so no kernel packet (q, p) can be subtracted. By uniqueness of the minimum demand it is $\mu(r + pm)$. The negative identity is symmetric, using the representative with $0 \leq x < q$.

By Lemma 2.3, between WRITEs the stack is a suffix of a canonical shortest-demand word and the mixed alternating part has uniformly bounded length. Equations (44)–(45) show that translating a deep residual tail changes only the number of complete unary m-packets. Thus a phase may be specified by the debt state, the bounded mixed suffix, the unary symbol and unary length modulo m, together with the residual class modulo pm or qm. Only finitely many phases occur. □

Choose once and for all

$$H \geq \max(p, q)$$

large enough that the finite symbol-level quotients of Lemma 8.1 are homogeneous whenever $\rho > H$ or $\rho < -H$. In the positive tail, feeding sufficiently many 1's first consumes the bounded mixed head and then accumulates positive mismatches on the unary zero tail until the positive outer-WRITE reset; for H large enough there are at least $h + 1$ unary zero cells available, and the residual only increases along this forcing word. The negative statement is symmetric, using a run of 0's. Thus every sufficiently deep phase can reach the sign-specific reset phase with positive probability without leaving its tail. Consequently, after transient phases are discarded, the finite phase kernel has a unique recurrent class. Restrict to that class and let P_+ and P_- denote the critical i.i.d. one-symbol phase kernels. For a bounded transition reward f, let

$$\bar{f}_\sigma(x) = \frac{q}{m} f(x, 0) + \frac{p}{m} f(x, 1), \qquad \sigma \in \{+, -\},$$

where $f(x, a)$ denotes the reward of the deterministic transition from phase x on symbol a. If π_σ is the invariant probability of the recurrent phase class, define

$$v_\sigma(f) = \pi_\sigma \bar{f}_\sigma.$$

The finite Poisson equation

$$g_\sigma - P_\sigma g_\sigma = \bar{f}_\sigma - v_\sigma(f) \tag{46}$$

has a bounded solution, unique up to an additive constant.

8.1 Bridge perturbation

Under the fixed-Parikh bridge, if the current residual is ρ_t, then

$$\mathbb{P}(X_{t+1} = 1 \mid \mathcal{F}_t) = \frac{p}{m} - \frac{\rho_t}{m(n - t)}. \tag{47}$$

Thus the deviation from the critical i.i.d. symbol law is

$$\delta_t = -\frac{\rho_t}{m(n - t)}.$$

For a fixed-Parikh bridge,

$$\operatorname{Var}(\rho_t) = pq \frac{t(n - t)}{n - 1}. \tag{48}$$

Therefore

Lemma 8.2 (Total bridge-kernel perturbation)**.**

$$\mathbb{E}\sum_{t=1}^{n-1}|\delta_t| \le \frac{\pi\sqrt{pq}}{2m}\sqrt{n}+O(1). \tag{49}$$

Proof. By Cauchy–Schwarz and (48),

$$\mathbb{E}|\delta_t| \le \frac{\sqrt{pq}}{m\sqrt{n-1}}\sqrt{\frac{t}{n-t}}.$$

Summing and comparing with

$$\int_0^1 \sqrt{\frac{x}{1-x}}\,dx = \frac{\pi}{2}$$

gives (49). □

8.2 The central band

Let

$$L_H = \#\{1 \le t \le n : |\rho_t| \le H\}.$$

Since $N_1(t)$ is hypergeometric with parameters (n, pg, t) and

$$\rho_t = mN_1(t) - pt,$$

a direct Stirling bound on the hypergeometric probability mass function gives, uniformly in admissible r,

$$\mathbb{P}(\rho_t = r) \le C_{p,q}\sqrt{\frac{n}{t(n-t)}}.$$

(The finitely many endpoint values of t are absorbed into the constant.) The band contains only finitely many residual values, hence

$$\boxed{\mathbb{E}L_H = O(\sqrt{n}).} \tag{50}$$

Because $H \ge \max(p,q)$, a path cannot jump directly from the deep positive tail $\{\rho > H\}$ to the deep negative tail $\{\rho < -H\}$ in one input step. Therefore the number of maximal deep-tail intervals is at most $L_H + 1$ up to an absolute endpoint constant.

8.3 Poisson decomposition under the bridge

At time t in a fixed deep sign, write Q_t for the bridge phase kernel. Since the next-symbol probabilities differ from the critical law only by δ_t,

$$\|(Q_t - P_\sigma)u\|_\infty \le 2\|u\|_\infty|\delta_t| \tag{51}$$

for every bounded phase function u, and likewise the bridge conditional reward mean differs from $\bar{f}_\sigma$ by at most $2\|f\|_\infty|\delta_t|$.

For an interior step of a deep-tail interval, combine (46) with Q_t to write

$$f_t - v_\sigma(f) = \big(g_\sigma(Y_t) - g_\sigma(Y_{t+1})\big) + \Delta M_t + \varepsilon_t, \tag{52}$$

where (ΔM_t) is a martingale-difference sequence with uniformly bounded increments and

$$|\varepsilon_t| \le C_{p,q,f}|\delta_t|. \tag{53}$$

Indeed, subtract first the bridge conditional mean of f_t, and then replace both the bridge reward mean and $Q_t g_\sigma$ by their critical counterparts; the two replacement errors are controlled by (51). The remaining centered terms form the martingale difference, while the g_σ terms telescope. The at most two transitions adjacent to each deep-tail interval endpoint are charged to the interval endpoint error.

Theorem 8.3 (Finite-to-macroscopic operational transfer)**.** *For every bounded operational reward f there is a constant $K_{p,q,f} < \infty$ such that*

$$\boxed{\mathbb{E}\,|C_g(f) - [nv_-(f) + (v_+(f) - v_-(f))A_g]| \le K_{p,q,f}\sqrt{n}.} \tag{54}$$

Proof. Compare f_t with $v_+(f)$ when $\rho_t > 0$ and with $v_-(f)$ when $\rho_t \le 0$. On the central band the absolute discrepancy is bounded by a constant times L_H, hence has expectation $O(\sqrt{n})$ by (50).

On every maximal deep positive or negative interval, sum (52). The g_σ terms telescope. Since g_σ is bounded, all interval endpoint and omitted boundary-step contributions have total L^1 size $O(\mathbb{E}(L_H + 1)) = O(\sqrt{n})$. The accumulated perturbation has expectation

$$O\left(\mathbb{E}\sum_{t<n} |\delta_t|\right) = O(\sqrt{n})$$

by Lemma 8.2. The sum of all martingale differences has bounded increments and predictable quadratic variation $O(n)$, so Cauchy–Schwarz gives L^1 norm $O(\sqrt{n})$.

Finally,

$$\sum_{t=1}^{n} \left[v_-(f) + (v_+(f) - v_-(f))\mathbf{1}_{\{\rho_t>0\}}\right] = nv_-(f) + (v_+(f) - v_-(f))A_g.$$

Combining the central, endpoint, perturbation, and martingale contributions proves (54). □

9 Macroscopic operational limit laws

The critical bridge invariance principle gives

$$\frac{\rho_{\lfloor nt \rfloor}}{\sqrt{pq\,n}} \Longrightarrow B^{\mathrm{br}}(t)$$

in $C[0,1]$, where B^{br} is a standard Brownian bridge. The zero set of a Brownian bridge has Lebesgue measure zero almost surely, so the positive-occupation functional is almost surely continuous at the limit. The positive occupation time of a Brownian bridge is uniform on $[0,1]$; uniform sojourn laws also admit exchangeable-increment and Lévy-bridge formulations [7, 8]. Hence

$$\frac{A_g}{n} \Longrightarrow U, \qquad U \sim \mathrm{Unif}(0,1). \tag{55}$$

Classical fixed-score ballot/election-return results provide a discrete counterpart to this occupation phenomenon [6].

Theorem 9.1 (Fixed-semantics operational limit)**.** *For every bounded operational reward f,*

$$\boxed{\frac{C_g(f)}{n} \Longrightarrow v_-(f) + U\big(v_+(f) - v_-(f)\big), \qquad U \sim \mathrm{Unif}(0,1).} \tag{56}$$

Proof. Divide (54) by n. The L^1 error tends to zero. Apply (55) and Slutsky's theorem. □

Corollary 9.2 (Uniform LIFO-obstruction law). *If $p \neq q$, then*

$$\boxed{\frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big).} \tag{57}$$

If $p = q = 1$, then

$$\frac{R_g}{2g} \longrightarrow \frac{1}{2}$$

in probability.

Corollary 9.3 (Rank-one macroscopic fluctuations). *Let $f_1, \dots, f_k$ be bounded operational rewards and*

$$\boldsymbol{C}_g = (C_g(f_1), \dots, C_g(f_k)).$$

Write

$$V_\pm = (v_\pm(f_1), \dots, v_\pm(f_k)).$$

Then

$$\frac{\boldsymbol{C}_g}{n} \Longrightarrow V_- + U(V_+ - V_-), \tag{58}$$

and

$$\boxed{\frac{\mathrm{Cov}(\boldsymbol{C}_g)}{n^2} \longrightarrow \frac{1}{12}(V_+ - V_-)(V_+ - V_-)^T.} \tag{59}$$

In particular the limiting covariance matrix has rank at most one.

Proof. Apply Theorem 9.1 componentwise; the same occupation variable U drives every coordinate. Since each normalized count is bounded, second moments converge. The covariance formula follows from $\mathrm{Var}(U) = 1/12$. □

Theorem 9.4 (Three-scale LIFO realization). *For the balanced fixed-Parikh random-ordering model,*

$$\boxed{\tau_g = O_{\mathbb{P}}(1), \qquad \frac{B_g}{\sqrt{n}} \Longrightarrow \mathrm{Rayleigh}\left(\frac{\kappa_{p,q}}{\sqrt{pq}}\right), \qquad \frac{R_g}{n} \Longrightarrow \mathrm{Unif}\big(\min(r_-, r_+), \max(r_-, r_+)\big)} \tag{60}$$

for $p \neq q$, with the last law degenerating to $1/2$ for $p = q = 1$.

10 Physical-lift duality

For a word w, let $\bar{w}$ denote the bitwise complement. Interchanging p and q and complementing input reverses the residual:

$$\rho_t^{q,p}(\bar{w}) = -\rho_t^{p,q}(w). \tag{61}$$

Moreover shortest demands are coordinate-swapped:

$$\mu_{q,p}(-r) = \mathrm{swap}(\mu_{p,q}(r)). \tag{62}$$

The two canonical words $\eta(x, y)$ and $\eta(y, x)$ have the same alternating prefix and opposite unary tails. By the canonical-segment property, outer WRITE cannot occur until the mixed prefix has disappeared. During that common mixed phase, complemented input produces opposite debt. The unary phases are then mirror images.

Theorem 10.1 (Physical-lift duality)**.** *For every finite word w,*

$$B_{p,q}(w) = B_{q,p}(\bar{w}), \qquad O_{p,q}(w) = O_{q,p}(\bar{w}). \tag{63}$$

Consequently, for every g,

$$(B_g, O_g)_{p,q} \overset{d}{=} (B_g, O_g)_{q,p}$$

under the corresponding balanced fixed-Parikh laws, and

$$\boxed{\kappa_{p,q} = \kappa_{q,p}.} \tag{64}$$

Proof. Couple the $p{:}q$ run on w with the $q{:}p$ run on $\bar{w}$. At the start of every WRITE episode the two canonical stacks have the same length, the same alternating-prefix length $2\min(x, y)$, and complementary unary tails. The boundary guard prevents an outer WRITE while that mixed prefix is still present, so both runs necessarily pop the entire mixed prefix. If v ones occur in the original input during those $2\min(x, y)$ steps, then the original debt is $v - \min(x, y)$, whereas the complemented run has debt $2\min(x, y) - v - \min(x, y)$; the two debts are therefore opposite exactly when the unary tails begin. From that point onward the visible unary symbols are complementary and the coupled input symbols are complementary, so MATCH/mismatch status agrees and the debts remain opposite. Hence a positive outer WRITE in one run is paired with a negative outer WRITE at the same time in the other. If no outer WRITE occurs, the equal-length stacks empty simultaneously, so the next input is a bottom WRITE in both runs. Induction over WRITE episodes proves equality of bottom- and outer-WRITE indicators at every time, yielding (63). The complement map is a bijection between $\mathcal{F}_{gq,gp}$ and $\mathcal{F}_{gp,gq}$. Finally, (64) follows from the bottom-WRITE Rayleigh scale or directly from the mirrored invariant measures. □

11 Near-diagonal universality

Let

$$p = q + 1, \qquad m = 2q + 1, \qquad h = q.$$

The central physical-lift fibers simplify sharply.

Proposition 11.1 (Near-diagonal central fibers)**.** *At block boundaries, for $0 \le z \le q$ the central semantic shell z has only the empty lift*

$$E_z = (D_z, \varnothing).$$

For $z = -r$, $1 \le r \le q$, the only central lifts are

$$E_{-r} = (D_{-r}, \varnothing), \qquad F_{-r} = (D_{q-r}, 1^m). \tag{65}$$

Proof. The proof is by most-recent-WRITE analysis. A positive outer WRITE has

$$\mu((q + 1)m) = (m, 0),$$

so it creates only a zero tail. Before the next WRITE, a nonempty block-boundary configuration on that tail has stack 0^{km} for some $k \ge 1$ and nonnegative debt, hence semantic shell at least $(q + 1)k \ge q + 1$. It cannot lie in the central range.

A negative outer WRITE has

$$\mu(-(q + 1)m) = (q - 1, 3q + 3),$$

so its mixed prefix has length $2(q-1) < m$ and its final tail is unary one. Let a later block-boundary suffix have length km and contain a zeros. Debt was reset to zero at the WRITE, and positive debt can subsequently be created only by one-over-zero mismatches that pop zero cells. Since the written word contained only $q-1$ zeros, the current debt satisfies

$$d \le (q-1) - a.$$

Hence its semantic shell obeys

$$z = a + d - qk \le q - 1 - qk.$$

If $k \ge 2$, then $z \le -q-1$, outside the central range. Thus a central nonempty suffix forces $k = 1$; since the mixed prefix is shorter than m, the surviving length-m suffix is exactly 1^m.

It remains to inspect bottom WRITEs. Both possible bottom-WRITE residuals have the form

$$r = jm + q, \qquad -q-1 \le j \le q.$$

If $\mu(r) = (x, y)$ and $\ell = x + y$, then

$$mx - q\ell = jm + q,$$

so $\ell \equiv -1 \pmod m$. Write $\ell = km - 1$. Solving gives

$$x = j + qk, \qquad y = k(q+1) - 1 - j.$$

For $-q \le j \le q$, the minimum is $k = 1$, so the newly written word has length $m-1$ and is completely consumed by the remaining $m-1$ symbols of the block. The single exceptional value $j = -q-1$ requires $k = 2$; it occurs exactly for input zero from E_{-q}, and

$$\mu(r) = (q-1, 3q+2).$$

Its canonical word is $(10)^{q-1}1^{m+2}$, so after the remaining $m-1$ pops the next block boundary contains precisely 1^m. These cases give exactly (65). □

Let

$$D_q = \sum_{r=1}^{q} \nu(F_{-r}).$$

Since every central semantic shell has total invariant mass one,

$$\kappa_{q+1,q} = 2q + 1 - D_q. \tag{66}$$

Lemma 11.2 (Near-diagonal predecessor bound)**.** *If a block ends in one of the nonempty central lifts F_{-r}, then the preceding semantic shell satisfies*

$$Z_j \le -q.$$

Proof. If the block contains a WRITE, consider its last WRITE. A positive outer WRITE cannot produce a one-tail. A negative outer WRITE inserts a word of length $2m$ before its surviving suffix, so with at most $m-1$ input symbols left in the block there are too few subsequent pops for the final stack to be exactly 1^m. A bottom WRITE can occur only on the first symbol of a block by Lemma 5.1, and the only such case that can end in a nonempty central lift is the transition from E_{-q}. It remains to consider a block with no WRITE. Then every one of the m input steps pops an old cell, so the starting stack has the form $T1^m$ with $|T| = m$. Let a be the number of zeros in this stack, and let x_0 be the zero-count of the canonical word produced by the most recent WRITE. Because at least m trailing ones survive and the demand is kernel-free, $x_0 \le q-1$. Positive debt can have increased only by popping zero cells through one-over-zero mismatches, so if the current debt is d then

$$d \le x_0 - a.$$

The residual shell of a length-$2m$ stack is

$$Z_j = a + d - 2q \le x_0 - 2q \le -q - 1.$$

This proves the claim. $\square$

Let

$$Y_q = K - (q+1), \qquad K \sim \mathrm{Bin}\left(2q+1, \frac{q+1}{2q+1}\right).$$

Then

$$\mathbb{E}Y_q = 0, \qquad \mathrm{Var}(Y_q) = \frac{q(q+1)}{2q+1}.$$

By Lemma 11.2 and unit shell mass,

$$\begin{aligned} D_q &\le \sum_{z \le -q} \mathbb{P}\{z + Y_q \in [-q, -1]\} \\ &\le \mathbb{E}\big[\min(Y_q+1, q)\mathbf{1}_{\{Y_q \ge 0\}}\big] \\ &\le 1 + \mathbb{E}Y_q^+ \\ &\le 1 + \frac{1}{2}\sqrt{\frac{q(q+1)}{2q+1}}. \end{aligned} \tag{67}$$

Theorem 11.3 (Near-diagonal universal Rayleigh scale)**.** *For $p = q+1$,*

$$\boxed{\frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \longrightarrow 2.} \tag{68}$$

More quantitatively,

$$2\sqrt{\frac{q}{q+1}} - \frac{1}{2\sqrt{2q+1}} \le \frac{\kappa_{q+1,q}}{\sqrt{q(q+1)}} \le \frac{2q+1}{\sqrt{q(q+1)}}. \tag{69}$$

Consequently, after first taking $g \to \infty$ and then $q \to \infty$,

$$\frac{B_g}{\sqrt{(2q+1)g}} \Longrightarrow \mathrm{Rayleigh}(2).$$

Proof. Combine (66) with (67) and divide by $\sqrt{q(q+1)}$. Both sides of (69) tend to two. The final statement follows from Theorem 6.2. $\square$

12 Finite physical-lift kernels and explicit computation of $\kappa_{p,q}$

The distributional results above require only the invariant-measure definition of $\kappa_{p,q}$. We now explain why the same constant is effectively computable from a finite matrix-kernel system.

12.1 Affine shortest-demand tails

At block boundaries the shortest demands have explicit forms. For $z \geq 0$,

$$\mu(mz) = \left(z + q\left\lceil \frac{z}{p} \right\rceil, p\left\lceil \frac{z}{p} \right\rceil - z\right). \tag{70}$$

For $n \geq 0$,

$$\mu(-mn) = \left(q\left\lceil \frac{n}{q} \right\rceil - n, n + p\left\lceil \frac{n}{q} \right\rceil\right). \tag{71}$$

Hence

$$\mu(m(z+p)) = \mu(mz) + (m, 0), \tag{72}$$

$$\mu(-m(n+q)) = \mu(-mn) + (0, m). \tag{73}$$

Thus every p positive semantic shells add one unary packet 0^m, while every q negative shells add one unary packet 1^m.

Proof of (70)–(73). For $mz > 0$, all integer solutions of $px - qy = mz$ are

$$x = z + qk, \qquad y = pk - z.$$

The least k making both coordinates nonnegative is $k = \lceil z/p \rceil$, giving (70). The negative formula is symmetric. The affine identities follow by replacing z by $z + p$ and n by $n + q$. □

By Lemma 2.3, a reachable stack is a suffix of a canonical shortest-demand word. The mixed alternating head has bounded length because every freshly written shortest demand has minority count at most h; the only unbounded part is a unary tail. Write the unary-tail length as

$$L = m\ell + r, \qquad 0 \leq r < m.$$

Modulo complete unary packets a^m, the remaining information consists of the debt state, the unary symbol, the bounded mixed head, and r. There are finitely many possibilities.

Theorem 12.1 (Finite physical-lift kernel)**.** *For every coprime p, q, there exist finite positive- and negative-tail phase sets and finite families of rational matrices $\{M_j^+\}$ and $\{M_j^-\}$ such that the invariant shell vectors $v_\ell^\pm$ satisfy, for all sufficiently deep levels,*

$$v_\ell^\pm = \sum_j v_{\ell-j}^\pm M_j^\pm. \tag{74}$$

The corresponding matrix Laurent polynomials

$$\mathcal{K}_\pm(\lambda) = I - \sum_j M_j^\pm \lambda^j \tag{75}$$

have rational coefficients. Together with finitely many central balance equations and the normalization $\nu(o) = 1$, the bounded nonnegative solutions of (74) *uniquely determine $\kappa_{p,q}$.*

Proof. The affine identities (72)–(73) show that translation in a deep semantic tail adds only complete unary m-packets. The canonical-segment property bounds the mixed head independently of level. Quotienting the unary length modulo m therefore leaves finitely many physical phases and one integer level. During one m-symbol input block, the level can change by only a bounded amount, and once the tail is sufficiently deep the transition probabilities depend only on the phase and the level increment, not on the absolute level. Under the critical law every block word has rational probability $(p/m)^k(q/m)^{m-k}$, so the matrices are rational. The invariant equation in the homogeneous tail is exactly (74). Irreducibility and the normalization $\nu(o) = 1$ make the recurrent invariant measure unique, hence the central empty-state coordinates and $\kappa_{p,q}$ are uniquely determined. □

This is a finite-phase, level-homogeneous matrix-analytic structure of the same general type that underlies quasi-birth-and-death and matrix-geometric methods [10]. The matrix-analytic machinery itself is classical; the point here is that the shortest-demand pushdown quotient canonically produces such a kernel whose central boundary values encode physical LIFO rematerialization.

12.2 Small explicit kernel examples

The two smallest asymmetric $q = 1$ cases reduce to scalar tail recurrences. We record their eliminations as compact reproducibility checks, and then give the first genuinely multi-phase example at 3:2.

The ratio 2:1. In negative shell $-n$ there are two deep phases,

$$A_n = (D_{-1}, 1^{3(n-1)}), \qquad B_n = (D_0, 1^{3n}).$$

Write $a_n = \nu(A_n)$, $b_n = \nu(B_n)$ and $u_n = a_n - b_n$. The unit-shell identity gives $a_n + b_n = 1$. Exact enumeration of the eight critical three-symbol blocks gives

$$u_{n-2} - 6u_{n-1} + 39u_n - 8u_{n+1} = 0 \tag{76}$$

in the homogeneous tail, while the finite central balance equations are

$$31u_1 - 8u_2 = 15, \qquad 4u_1 - 39u_2 + 8u_3 = 3. \tag{77}$$

The characteristic polynomial is

$$Q_{2,1}(R) = 8R^3 - 39R^2 + 6R - 1.$$

It has a unique real root $R > 1$. Boundedness of the invariant lift profile cancels that mode, which is equivalent to

$$u_1(31R^2 - 4R + 1) = 15R^2 - 3R.$$

With $\theta = (1 + u_1)/2 = a_1$, elimination of R between this relation and $Q_{2,1}(R) = 0$ gives

$$8\theta^3 - 3\theta - 1 = 0. \tag{78}$$

Thus

$$\kappa_{2,1} = 2 + \theta = 2.737843258897860\ldots. \tag{79}$$

The ratio 3:1. The same two negative phases occur with unary packet length four. Writing again $u_n = a_n - b_n$, exact sixteen-block enumeration gives

$$81u_{n+1} - 364u_n + 54u_{n-1} - 12u_{n-2} + u_{n-3} = 0, \tag{80}$$

and the finite central equations are

$$283u_1 - 81u_2 = 148, \tag{81}$$

$$-27u_1 + 364u_2 - 81u_3 = -40, \tag{82}$$

$$9u_1 - 54u_2 + 364u_3 - 81u_4 = 4. \tag{83}$$

The characteristic polynomial is

$$Q_{3,1}(R) = 81R^4 - 364R^3 + 54R^2 - 12R + 1.$$

Cancelling its unique unstable real mode gives

$$u_1(81R^3 - 81R^2 + 27R - 3) = 148R^2 - 40R + 4.$$

Eliminating R and setting $\theta = (1 + u_1)/2$ yields

$$54\theta^4 - 18\theta^2 - 8\theta - 1 = 0, \tag{84}$$

so

$$\kappa_{3,1} = 2 + \theta = 2.750694578264632\ldots. \tag{85}$$

The ratio 3:2. Here the deep negative tail has seven phases. Writing the negative semantic shell as $-n$, the four odd-shell phases for $n = 2j + 1$ are

$$\begin{aligned}
O_0(j) &= (D_{-1}, 1^{5j}),\\
O_1(j) &= (D_0, 0\,1^{5j+4}),\\
O_2(j) &= (D_0, 10\,1^{5j+3}),\\
O_3(j) &= (D_1, 1^{5(j+1)}),
\end{aligned}$$

and the three even-shell phases for $n = 2j$ are

$$\begin{aligned}
E_0(j) &= (D_{-2}, 1^{5(j-1)}),\\
E_1(j) &= (D_{-1}, 0\,1^{5j-1}),\\
E_2(j) &= (D_0, 1^{5j}).
\end{aligned}$$

Enumerating the 32 five-symbol blocks with their exact critical probabilities $3^k 2^{5-k}/5^5$ gives a rational seven-phase Laurent kernel whose determinant factors as

$$\det \mathcal{K}_{3,2}(\lambda) = \frac{(\lambda - 1)^2 C_3(\lambda) K_5(\lambda)}{5^{20}\lambda^6}, \tag{86}$$

where

$$\begin{aligned}
C_3(\lambda) &= 59049\lambda^3 - 1531872\lambda^2 + 9472\lambda - 1024,\\
K_5(\lambda) &= 59049\lambda^5 - 1387530\lambda^4 + 16632265\lambda^3\\
&\quad - 1448160\lambda^2 + 11520\lambda + 1024.
\end{aligned}$$

The critical phase kernel has invariant vector

$$\left(\frac{1}{4}, \frac{3}{50}, \frac{1}{10}, \frac{9}{100}, \frac{1}{4}, \frac{1}{25}, \frac{21}{100}\right) \tag{87}$$

in the order $(O_0, O_1, O_2, O_3, E_0, E_1, E_2)$. Coupling this exact tail kernel to the finite central balance equations gives numerically

$$\kappa_{3,2} = 4.64089572244120\ldots. \tag{88}$$

An independent direct probability propagation, extrapolated from block horizons 100 and 200, gives $4.6408956068\ldots$, independently confirming the first six decimal places. The rematerialization constant need not be rational even though the underlying transition probabilities are rational: already the cubic (78) has no rational root, so $\kappa_{2,1}$ is irrational.

Remark 12.2 (Algebraicity and regular kernels)**.** The finite kernels provide an exact rational boundary-value problem for $\kappa_{p,q}$. When the associated matrix Laurent pencil is regular, companion linearization shows that its characteristic modes are algebraic, and the finite central boundary equations then make the normalized bounded solution algebraic. This applies to the explicit kernels above. We do not require, and do not use, an unconditional algebraicity theorem for every coprime p, q in the probabilistic results of this paper.

13 Computational audit

Computation is used only as an independent transcription and boundary-case check; no computational result is a premise of the proofs above.

A direct exhaustive implementation of the centered transition schema was used to check the following finite statements.

1. For coprime $1 \leq p, q \leq 5$ and all balanced fibers with feasible $g \leq 3$, the number of obstruction-free balanced words is exactly 2^g.
2. The finite survival formulas (11)–(12) agree exactly with exhaustive enumeration in the same range.
3. The primitive SEARCH distribution (15) agrees with exhaustive enumeration for every coprime $1 \leq p, q \leq 7$.
4. Long critical-i.i.d. simulations for representative ratios 2:1, 3:1, 3:2, 4:3, and 5:2 agree with the explicit bulk rates (40)–(41) within ordinary sampling error.
5. Direct state-space exploration of the centered normalizer over ordered coprime pairs $1 \leq p, q \leq 20$ through breadth-first search depth 80 checks the state–stack residual invariant, two-cell qualitative visibility, transition totality, and zero-residual characterization; 1,909,191 reached configurations are checked in total across the 255 ordered coprime ratios.
6. For 3:2, an independent exact-rational reconstruction of the seven deep negative-tail phases reproduces the determinant factorization (86), the invariant vector (87), and the quoted value of $\kappa_{3,2}$ to six decimal places after an independent long-horizon extrapolation.
7. For 2:1 and 3:1, independent exact block enumeration reproduces the scalar tail characteristic polynomials, and symbolic resultants with the displayed central balance equations reproduce (78) and (84).

These checks are designed to detect sign, parity, and endpoint transcription errors. They are not substitutes for the symbolic arguments.

14 Related work and novelty boundary

Several ingredients used in this paper are classical and are not claimed as new.

Fixed-composition words and ballot occupation. Uniform fixed-Parikh words are the same objects as fixed-score ballot records or sampling-without-replacement bridges. Takács studied the number of proportional lead positions in such records [6]. At the continuum level, the positive occupation time of a bridge is uniform; Fitzsimmons and Getoor proved a corresponding result for Lévy bridges [8]. We use these facts to identify the macroscopic mixing variable, not as a novelty claim.

Local time and Rayleigh laws. Rayleigh laws for local-time/contact parameters of zero-drift lattice-path bridges are part of the analytic-combinatorics and fluctuation-theory literature; see, for example, Banderier and Wallner [9]. Our new quantity is not the scalar zero local time itself: a semantic primitive excursion carries a random physical empty-lift reward with mean $\kappa_{p,q}$, and that automaton-specific reward changes the Rayleigh scale.

Probabilistic pushdown models versus random input. Probabilistic pushdown automata traditionally place randomization in the transition or computation mechanism; see, for example, Hromkovič and Schnitger [3], and the recent probabilistic input-driven model of Rose and Okhotin [4]. The automaton studied here is deterministic. Probability enters only through a distribution on input orderings inside a fixed Parikh fiber, so the questions and observables are different from probabilistic acceptance or determinization.

Pushdown cost and Parikh semantics. Average stack-cost objectives for Büchi pushdown automata were studied by Michaliszyn and Otop [5]. Their cost is assigned to stack contents along an infinite accepting run. By contrast, our variables count concrete SEARCH and WRITE mechanisms of a deterministic finite-word normalizer under a fixed-semantic random-ordering law. Ganty and Gutiérrez study Parikh images of pushdown automata, where symbol ordering is explicitly forgotten [2]; here we hold the Parikh image fixed precisely in order to measure the cost of the forgotten ordering.

Matrix-analytic methods. Finite-phase level processes and matrix-geometric/QBD methods are classical [10]. We do not claim a new general matrix-analytic method. The contribution is the family-specific reduction from shortest-demand pushdown lifts to a finite physical-lift kernel whose boundary invariant gives $\kappa_{p,q}$.

Scope of the contribution. The main new theorem package is therefore the following synthesis:

$$\boxed{\begin{array}{c}\text{fixed shortest-demand semantics} \longrightarrow \text{ random order}\\ \longrightarrow \text{ quantified physical LIFO chronology}\end{array}}$$

with exact finite obstruction counts, a physical-lift invariant measure, a $p{:}q$-specific Rayleigh rematerialization law, an $O(\sqrt{n})$ operational transfer theorem, explicit uniform macroscopic LIFO limits, rank-one joint fluctuations, pathwise $p \leftrightarrow q$ duality, and near-diagonal Rayleigh universality. We make no absolute priority claim beyond this explicitly stated family-specific synthesis.

15 Conclusion

The $p{:}q$ shortest-demand quotient deliberately forgets normalization chronology: distinct physical state–stack configurations can represent the same semantic residual. By sampling uniformly inside a balanced fixed-Parikh fiber, this paper holds the semantics fixed and turns that lost chronology into a measurable random variable.

Three scales emerge. Direct realization typically fails after only $O(1)$ symbols. Physical rematerialization occurs on the diffusive $\sqrt{n}$ scale and has a Rayleigh law whose scale is the physical-lift invariant $\kappa_{p,q}/\sqrt{pq}$. SEARCH and outer-WRITE obstruction accumulate on the linear scale, where a single uniform bridge-occupation variable drives all bounded operational observables and forces rank-one macroscopic covariance.

The resulting picture is

$$\boxed{\text{semantic arithmetic} \;\longrightarrow\; \text{physical lift fibers} \;\longrightarrow\; \text{quantitative LIFO cost.}}$$

In this sense the normalizer provides a controlled model in which the semantics is fixed exactly and the cost of LIFO ordering can be isolated at finite, diffusive, and macroscopic scales.

Statements and Declarations

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Data and Code Availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, formulas, and proofs needed for the mathematical results are contained in the manuscript. No external dataset or executable code is required to establish the theoretical results. Computational consistency checks summarized in Section 13 are auxiliary validation tools and do not replace the symbolic proofs.

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